

Psychophysiological Predictors of Soldier Performance in Tunnel Warfare: A Field Study on the Correlates of Optimal Performance in a Simulation of Subterranean Combat

Corinne Berger, PhD^{ID}*; Uzi Ben-Shalom, PhD^{*}; Niv Gold, MD[†]; Avishai Antonovsky, PhD^{ID}[†]

ABSTRACT Tunnel operations produce unique psychophysiological activation that is correlated with cognitive impairment and lower performance. This study introduces a new concept: subterranean operational potential (SOP) and assesses its psychophysiological correlates for performance prediction in underground spaces. 138 soldiers of elite infantry battalions, with/without previous experience, who participated in a simulation of tunnel warfare. Physical, psychological, cognitive style, and performance measures were collected. SOP has three sub-components: performance, leadership, and orientation. Leadership and performance both were negatively correlated with perceived stress. Claustrophobia was negatively correlated with leadership. The cognitive style was positively correlated with performance. Saliva cortisol levels were significantly higher before the simulation. Inexperienced and experienced differed in the change in before-after saliva cortisol levels.

INTRODUCTION

Underground warfare includes the establishment of underground facilities for defense, attack, and illegal border-crossing. Previous examples of underground combat include trench warfare in WWII and urban and jungle warfare in the Vietnam war. Such operations are an inevitable part of modern fighting with mega-cities becoming arenas for counterinsurgency operations in current and near-future security challenges. In October 2013, more than 30 Hamas-built tunnels were exposed, entering Israeli territory from the Gaza strip.¹ Following this discovery, the IDF began training soldiers for combat in this arena. IDF elite infantry battalions are now required to undergo training in subterranean conflict as part of their standard training.² Nevertheless, an empirical first-hand study of the psychophysiological dimensions of subterranean combat is scant.³

Examples of external attributes of underground spaces include low ceilings, steep slopes, uneven flooring, holes, oxygen deficiency, and little to no light.^{2,4} Importantly, subterranean environments include hazards unique to this specific operating environment.⁵ Although the objective physical, technological, and geographic aspects of subterranean warfare are receiving ample focus, the human aspect of this environment is insufficiently addressed.⁶ In fact, underground operations have been shown to produce increases in blood lactate, oxygen saturation, heart rate, and perceived stress exertion rate.⁷

The maze-like, non-geometrical aspects of underground tunnels require a unique cognitive ability to navigate and orient in this arena.³ Considering that psycho-physiological activation is correlated with decreased cognitive and memory

performance, an examination of the physiological, psychological, and cognitive correlates of operations in underground tunnels is greatly needed.⁷ To our knowledge, the unique cognitive characteristics essential for optimal functioning in this arena have not been adequately studied or defined in Israel or abroad.³ Examples of characteristics that contribute to optimal performance in this arena include, among others, an ability to perform in isolation, after having been separated from the group/chain of command, an ability to function in small, stifling spaces, and good technical ability.³

Presently, any existing knowledge regarding the characteristics contributing to good performance in this arena is “borrowed” from other combat environments.^{8–10} Consequently, the current study aimed to assess and identify physiological, psychological, and cognitive attributes of the unique psychological characteristics of tunnel operations.

To this end, we performed a two-stage study. The first study included qualitative in-depth interviews qualitative interviews with experienced combatants who operated inside tunnels.⁷ The preliminary goal of the qualitative study was to identify key factors of everyday experiences and individual variables (unique operational and leadership abilities) that would predict soldier success in subterranean warfare.

The interviews were conducted using a semi-structured interview method in which the same set of questions were presented to the various interviewees. First, the interviewees were asked to describe the chronology of their military service, and it was emphasized that the interview refers to personal experiences only and that they are not requested to divulge any classified information. They were then asked to describe an event that occurred during their service that they perceived as significant. Interviewees were also asked to describe a combat event in which they participated. The interviewees were asked to describe a person who functioned well in the subterranean environment and, alternatively, one who had difficulty in this environment. We explained to the interviewees that the aim of this qualitative study is to determine

*Sociology and Anthropology Department, Ariel University, Ariel, Israel

[†]Medical Corps, Israeli Defense Forces, Israel

doi:<https://doi.org/10.1093/milmed/usab312>

© The Association of Military Surgeons of the United States 2021. All rights reserved. For permissions, please e-mail: journals.permissions@oup.com.

personality traits that contribute to effective functioning or that may be detrimental to operating in this environment.

Content analysis revealed similarities in interview content regarding traits necessary or detrimental to soldier performance in underground combat. The following are the features that were found to be important for a fighter in a subterranean environment. They included traits such as an ability to function in small-confined spaces, coping in situations of extreme stress, ability to function independently even if all contact is lost between the soldier and the rest of the team, and an ability to function even after orientation loss (complete darkness/unknown depth or distances from point of entrance).

We proposed a unique theoretical construct that is a soldier's subterranean operational potential (SOP). To our knowledge, the subterranean environment is still being considered as a subtype of urban warfare. There have been no attempts to address subterranean spaces as a unique environment, to identify variables predicting operational capabilities in this arena, or to develop methods to assess these traits in the selection processes of potential subterranean fighters.

Based on the data collected from the interviews and previous research indicating resilience and agility as important to the performance of soldiers and junior leaders in ground combat,^{10,11} we concluded that the SOP construct is broken down into three components: (1) SOP orientation, a soldier's ability of remaining an ability navigate and maintain, (2) SOP performance, a soldier's potential for high operability in the subterranean arena, and (3) SOP leadership, a soldier's potential of becoming an effective leader in this arena. In general, the differentiation between officers and other ranks in the IDF occurs naturally through an ongoing selection process out of the entire cohort of Israeli conscripts. Consequently, we aimed to identify soldiers who possess the unique leadership qualities that meet the demands of subterranean combat. We then established a self-report questionnaire scale to measure the three SOP subcomponents.

In the second stage of the study, we endeavored to establish and validate an overall assessment battery to predict soldier SOP. Empirical studies indicated a cognitive style that contributes to better performance in similar settings: the field independent (FI) cognitive style.^{12,13} Cognitive styles are stable constructs, independent from general intellectual ability, that determine how people search for, organize, and process information in the environment.^{14,15} Field dependence (FD) individuals tend to maintain the structure of the field as it is presented while relying on external cues to process information.¹⁶ Field-independent individuals tend to impose their own structure on the field and rely on internal cues, separating essential information from its environment.¹⁷ Field independence has been researched among members of elite military operations units (i.e., bomb disposal experts and pilots) is the FD/ FI model of cognitive styles.^{12,13} Field dependence /field independence is assessed using the Mental Rotation Test paradigm.¹⁸ Research indicates that an FI cognitive style is fundamental to the restructuring of environmental information

into a global and flexible representation of the environment. For example, FI individuals performed better than FD individuals in a mapping task and in a spatial orientation and perspective-taking test, suggesting that FD affects learning and recovery from the navigation environment.¹⁹

Tunnel warfare requires the soldier to operate in small, constricting spaces often characterized as claustrophobic. As such, psychological traits relevant to this combat arena are the ability to withstand being in constricting spaces. These, in turn, lead to enhanced cognitive performance. Surprisingly, the effects of claustrophobia among military personnel have only been researched in the context of protective clothing (i.e., biological /chemical warfare) submarine operations.^{19–21} Moreover, most research conducted among military personnel examined claustrophobia as an outcome (dependent variable) rather than a precursor (independent variable) to performance.²² Despite the obviously claustrophobic characteristic of tunnel warfare, claustrophobia-related anxiety is not being readily assessed among soldiers expected to operate in such constricting spaces.

A measurable physiological response to stress includes the activation of the hypothalamus-pituitary-adrenal (HPA) axis. The HPA axis produces a cascade of endocrine events causing the secretion of the glucocorticoid hormones, most predominantly cortisol in humans.²³ As such, cortisol is usually referred to as the "stress hormone".²⁴ While most (85%-95%) of the blood cortisol is bound to proteins, a small portion of the cortisol, 5%-15%, is unbound and is therefore small enough to enter cells and body fluids such as saliva.²⁵

Salivary cortisol is considered the gold-standard stress biomarker in clinical sciences and psychology, allowing researchers to objectively measure a subjective response to a psychological stressor.²⁶ A review of studies of cortisol measures among submariners has shown them to have a higher tendency toward hypercortisolism and addressed the possible influences of elevated cortisol levels on their health and performance of submarine personnel.²⁷ Cortisol is also influenced by irregularities in the natural sleep-wake cycle (circadian rhythm) commonly experienced by operational military personnel.²⁸ Consequently, salivary cortisol provides essential objective complementary information to participants' self-assessment ratings.²⁶

Studies indicate that experience modulates the association between an emotional response and cortisol reactivity.²⁶ For example, while an anticipatory anxiety response was elicited in both experienced and non-experienced pharmacy and biotechnology students, non-experienced students showed a lower habituation response in both subjective and objective stress measures than experienced students at the end of the laboratory exercise.²⁶ An exaggerated physiological stress response leads to negative impacts on cognition.²⁹

Despite its importance, the topic of subterranean combat attributes is not adequately addressed in the scientific literature. Any existing studies have attempted to implement variables known to predict soldier performance in open terrain³⁰

adapting these variables to the contingencies of subterranean environment. For a comprehensive examination of these characteristics, we established a test battery including a self-report scale to assess SOP leadership, performance, and orientation and their correlates with self-reported claustrophobia and stress, a test of field-dependent/independent cognitive styles, and before and after salivary cortisol levels. Our research hypotheses were as follows.

Study Hypotheses

- (1) Perceived stress and claustrophobia would be negatively correlated with SOP leadership, SOP performance, and SOP orientation.
- (2) Field-independence would be positively correlated with the SOP leadership, SOP performance, and SOP orientation.
- (3) Total saliva cortisol would be higher in inexperienced soldiers compared to experienced soldiers.

METHOD

Study Design

A prospective empirical study design. All procedures and measurements were approved by the medical ethics review board and explained before data collection to all relevant military personnel, who gave voluntary written informed consent in accordance with the Declaration of Helsinki. The study examined a subset of IDF active duty soldiers belonging to elite reconnaissance infantry battalions. All were undergoing training of Military Subterranean Operations between February and December 2019.

Participants

One hundred thirty-eight regular service soldiers were recruited from reconnaissance battalions of the regular infantry brigades.

Recruitment Process

Data from nine groups were collected. The study was presented 2 days before participant recruitment. In accordance with conditions contained in HC 2.0716 and HQ 100.013, the study was presented without the presence of commanding officers. The soldiers were told that this is a study on the personal characteristics of soldiers performing in underground spaces and that, as part of the study, they would be asked to complete a set of pen-and-paper questionnaires and would be requested to give two saliva samples, before and after the simulation exercise. It was also emphasized that their participation would be entirely voluntary and that they may terminate their participation in the study without consequences. Soldiers were encouraged to ask questions, and any misunderstandings were clarified. Soldiers were told that they would be given additional opportunities to ask questions should any misunderstandings arise.

The soldiers received the informed consent form 2 days before recruitment and were encouraged to consult it privately and in their free time, and to consult with friends and family on the subject. On the day of the study, a short personal conversation was held between each soldier and the investigator to allow soldiers an opportunity to refuse to participate without peer pressure.

Participants

One hundred thirty-eight regular service soldiers recruited from reconnaissance battalions of the regular infantry brigades. The average age of participants in years was 20 years ($SD = 1$). Average seniority in months in military service was 14 years, $SD = 7$. 67% had rank of private and corporal, 33% were sergeants and 1st sergeants. Only 14% took part in a combat operational event.

Materials

Self-report scale on SOP

A nine-item questionnaire. Participants are requested to rate statements pertaining to their performance in the simulation exercise. Answers are rated on a 7-point Likert scale. Soldiers are requested to respond between 1 (not at all) to 7 (very much so). For example, subjects read the statement “I believe I am suitable for subterranean combat settings”. Two of the test questions must be reversed.

The Perceived Stress Scale-ten item version (PSS-10)

The scale includes items on a 5-point Likert scale ranging from zero to four. Four of the items are scored in the positive direction, while the other six must be reversed. Once these items are reverse-scored, higher overall total scores indicate a higher level of perceived stress (Cronbach's $\alpha = 0.810$).

Claustrophobia

Subscale of the Structured Clinical Interview for DSM Disorders.³¹ It includes six items from the specific-phobia section that are used for the diagnosis of claustrophobia. Researchers argue that while claustrophobia is sometimes not expressed in functional impairment, it may still be diagnosed.³² This point is of crucial importance. That is, although an individual may not report claustrophobia-related impairments in everyday life, these cannot be avoided in subterranean settings. Participants were defined as having claustrophobia if they rated “yes” on items 1, 2, 3, and 4 or 5 or above 5 on item 6.

Revised-Mental Rotation Test

This is a 24-item problem set. Each problem has a target figure shown on the left and four stimulus figures on the right. Two of these stimuli are rotated versions of the target figure, and the other two cannot be matched to the target figure. Respondents are required to “mentally” rotate the figures both around the vertical and the horizontal axes to determine which figures

match the target figure. One point is allocated for each correct answer, and a point is subtracted for each incorrect answer yielding a maximum of 48 points.

Cortisol

Measuring cortisol level in saliva is an indication of the stress response. The small-sample saliva collection was kept refrigerated until transferred to the Endocrinology Institute at Shamir Medical Centre laboratory, where tests were performed.

Demographic and military characteristics

A self-report questionnaire with the following covariates: age (in years), place of birth, educational attainment (in years), military role (fighter, medic, sniper, etc.), rank, previous tunnel warfare training (yes vs no), and previous combat experience.

Procedure

The underground simulation exercise: the simulation was conducted in accordance with actions occurring in actual subterranean theatres of operations. As part of the exercise, soldiers had to deal with an uncertain number of enemies armed with Kalashnikov rifles, concealed knives, and improvised explosive device.

After giving written consent, participants filled out questionnaires in a classroom setting, a day before the tunnel simulation exercise. On the day of the simulation exercise, participants gave the first saliva sample before proceeding to the tunnel simulation exercise. After the simulation exercise, they gave the second saliva sample and filled in the SOP questionnaire.

All participant data (demographics, self-report scales, field-dependent/independent cognitive styles test, and salivary cortisol sample) were coded by a unique random identification number. Data were stored anonymously and only the principle investigator had access to the participant data log that was kept as a secure file.

Participants were equipped with boots, tactical fighting load carriers, body-armor, M-16, shackles, and a backpack with 12 kg to simulate the equivalent weight of the task force for this type of operation.

Data Analysis

Statistical analyses were performed using SPSS 21 software (SPSS Inc.). Before conducting any parametric tests, the normality assumption of each dependent measure was verified using the Kolmogorov-Smirnov test. Asymmetric tests for the nominal demographic variables and correlation analyses were performed to compare study variables: saliva cortisol, stress perception scale, claustrophobia questionnaire, mental rotation test, and the three factors of the SOP index: performance, leadership, and orientation difficulty. Results were considered significant at a significance level of 95%.

RESULTS

Initial component analysis with Varimax rotation was performed, yielding a three-factor solution with Varimax rotation (explaining 73% of the variance): SOP performance (four items, Cronbach's $\alpha = 0.90$), SOP leadership (three items, Cronbach's $\alpha = 0.71$), and SOP orientation (Pearson's $r = 0.59$, $P < .001$).

SOP leadership scores were positively associated with SOP performance scores ($r = 0.601^{***}$, $P = .000$), while SOP orientation was uncorrelated with SOP leadership or SOP performance. Descriptive statistics and inter-correlations between research variables are provided in Table I.

As expected, Perceived Stress Scale scores (PSS) were positively correlated with claustrophobia scores ($r = 0.281^{**}$, $P = .007$). We hypothesized that SOP leadership, performance and orientation would be negatively correlated with stress and claustrophobia. These hypotheses were partly proven: SOP leadership and SOP performance were both negatively correlated with PSS scores ($r = -0.294^{**}$, $P = .006$; $r = -0.249^*$, $P = .021$). SOP leadership was negatively correlated with claustrophobia scores ($r = -0.363^{**}$, $P = .000$). Neither SOP performance nor SOP orientation was correlated with claustrophobia.

We hypothesized that a Mental Rotation Test (MRT) scores would be positively correlated with SOP leadership, SOP performance, and SOP orientation. This hypothesis was partially supported. MRT scores were positively correlated with SOP performance ($r = 0.232^*$, $P = .03$) but not SOP leadership or SOP orientation.

TABLE I. Descriptive Statistics and Correlation Analysis of Variables among All Participants

Variable	<i>n</i>	Mean	SD	1	2	3	4	5	6
1. CSS	111	0.427	1.073	–	0.281 ^{**}	0.109	–0.363 ^{**}	0.008	–0.025
2. PSS	94	2.221	0.575	–	–	0.007	–0.294 ^{**}	–0.249 [*]	–0.162
3. MRT	108	0.738	0.23	–	–	–	0.102	0.232 [*]	0.074
4. SOP leadership	108	5.162	1.343	–	–	–	–	0.601 ^{**}	0.160
5. SOP performance	107	5.604	1.062	–	–	–	–	–	0.066
6. SOP orientation	107	4.79	1.525	–	–	–	–	–	–

Abbreviations: CSS, Claustrophobia Stress Subscale; MRT, Mental Rotation Test; PSS, Perceived Stress Scale; SOP, Subterranean Operational Potential.

* $P < .05$.

** $P < .01$.

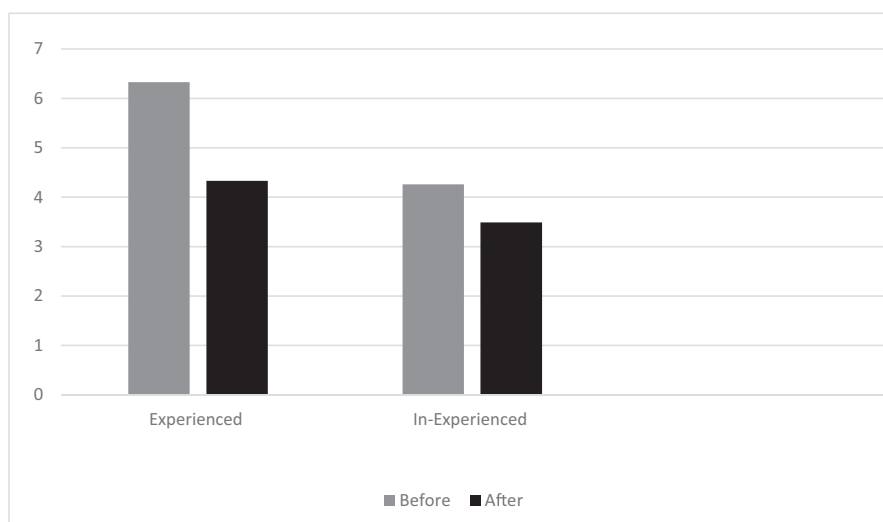


FIGURE 1. Cortisol level before and after tunnel warfare simulation among the level of experience.

TABLE II. Hierarchical Regression Analysis Predicting SOP Subscales

Variable	SOP leadership			SOP performance			SOP orientation		
	B	r^2	Δ	β	r^2	Δ	β	r^2	Δ
Step 1	Experience	0.03	-0.007	0.10	0.004		-0.09	0.001	
Step 2	Experience	0.07	0.12	0.10	0.08	0.076	-0.11	0.01	N.S.
	CSS	-0.27**		0.06			0.1		
	PSS	-0.17*		-0.19*			-0.17		
	MRT	0.13		0.24**			0.06		

Abbreviations: CSS, Claustrophobia Stress Subscale; MRT, Mental Rotation Test; PSS, Perceived Stress Scale; SOP, Subterranean Operational Potential.

* $P < .05$.

** $P < .01$.

We hypothesized that inexperienced soldiers would show significantly higher total salivary cortisol measures compared with experienced soldiers. This hypothesis was not proven, and no significant differences were found between overall saliva cortisol levels of experienced and inexperienced soldiers. Instead, there was a significant drop in saliva cortisol in the “after-simulation” sample compared to the before “before-simulation” sample of experienced soldier. This drop was not evident in the inexperienced group. (Wilks’ Lambda = 3.49, $P < .05$). The differences are presented in Figure 1. Saliva cortisol levels before and after the simulation were positively correlated ($r = 0.39^{**}$), and Saliva cortisol levels of all study participants were significantly higher before the simulation exercise ($t_{(105)} = 2.464$, $P = .015$).

Following our preliminary results, we computed a series of hierarchical regression analyses predicting the three SOP sub-scales. The level of experience was controlled in the first step and the predictors were Claustrophobia, Stress, and MRT scales. We found a significant level of r^2 (adjusted) for SOP leadership and SOP performance (r^2 (adjusted) = 0.11, $F = 5.11$, $P < .01$ and r^2 (adjusted) = 0.08, $F = 3.74$, $P < .01$, respectively). The result of the regression equation predicting

SOP orientation was not significant (r^2 (adjusted) = 0.01 N.S). The results are provided in Table II.

The results indicate that by controlling for previous experience in subterranean simulation, the variables significantly predict SOP leadership and performance. Stress level predicted both subscales ($\beta = -0.17$ and -0.19 , $P < .05$, respectively). Claustrophobia predicted SOP leadership ($\beta = -0.27$, $P < .001$) and MRT predicted SOP performance ($\beta = 0.24$, $P < .001$).

DISCUSSION

The current research directly addressed the question of the adaptation of combatants to subterranean environments. As hypothesized, subjective claustrophobia and stress measures were indicative of reduced performance in the underground simulation, with a positive correlation between claustrophobia and stress. FI cognitive style was positively correlated with SOP performance. That is, persons displaying FI cognitive styles are expected to display better performance in tunnel warfare. The current data are in accordance with a meta-analysis on the characteristics of successful military divers that indicated a strong sense of autonomy to be an important trait.³³

Our findings regarding the role of previous experience are in line with that of Redondo-Flórez et al.²⁶ who showed that prior experience did not extinguish the stress response but rather altered the individual's engagement of the HPA axis. Cortisol has been shown to be substantially released upon threatening psychological stressors and is an objective (di-) stress measure in critical incident training and operations.³⁴ As such, our findings are in line with a study that revealed an unexpected cortisol decrease in response to "positive stress" (i.e., eustress).

In various military frameworks (e.g., Air Force and Navy), dedicated tools have been developed to identify the psychological characteristics that predict optimal functioning in these combat environments. For the underground combat environment, no empirically assessed means for identifying relevant psychological characteristics have been developed. The likelihood of first-hand experience in underground infrastructures is scant because of the post-heroic style of warfare characterized by a decrease in citizens willing to fight and a rise in risk aversion. As a result, the importance of classification and development using live simulations is crucial. These stress-inducing measures ought to be designed to simulate the unique psychological demands of tunnel operations.

CONCLUSION

This study is focused on the un-developed topics of psychophysiological, cognitive, and behavioral elements of soldiers and leaders in tunnel warfare. The purpose of this study was to develop an objective tool for examining the mental potential of soldiers and leaders for operations in underground settings. We believe that the results of this study draw research attention to the mental fitness variable in subterranean warfare and identify the unique requirements of tunnel combat. The weaknesses of the study include a relatively small study sample and the use of subjective assessment tools. Future research may validate specific training programs designed for underground combat using a larger sample and additional indicators of successful performance.

ACKNOWLEDGMENT

We would like to thank Dr Reuven Gal for his valuable contribution to the preparation of this article.

FUNDING

The study was funded by the Medical Corps, Israeli Defense Forces, Grant number: 1827-2017.

CONFLICT OF INTEREST STATEMENT

We have no conflicts of interest to disclose.

REFERENCES

1. Watkins NJ, James AM: Digging into Israel: the sophisticated tunneling network of Hamas. *J Strateg Secur* 2016; 9(1): 84–103.
2. Shapir YS, Perel G: Subterranean warfare: a new-old challenge. In: Kurz A, Brom S, eds. *The lessons of operation protective edge*. Tel Aviv 2014; 51–57.

3. Stern N, Ben-Shalom U, Gold N, Berger C, Antonovsky A, Peleg D: Tunnel operations in the Israel Defense Forces: adapting the warrior ethos to post-heroic conflict. *Armed Forces Soc* 2020; 1–21.
4. Rossiter A: Bots on the ground: an impending UGV revolution in military affairs? *Small Wars and Insur* 2020; 31(4): 851–73.
5. School UAI: Subterranean operations. US Army Techniques. Department of the Army; 2018.
6. Bartolomeu JP: Operations in subterranean systems: Terrain and weather variable. *Secur Defen Q* 2020; 29(2): 39–60.
7. Tornero-Aguilera JF, Clemente-Suárez VJ: Effect of experience, equipment and fire actions in psychophysiological response and memory of soldiers in actual underground operations. *Int J psychophysiol* 2018; 128: 40–46.
8. Dekel R, Levinstein Y, Siegel A, Fridkin S, Svetlitzky V: Secondary traumatization of partners of war veterans: the role of boundary ambiguity. *J Family Psychology* 2016; 30(1): 63.
9. Rottman GL: *Tunnel Rat in Vietnam*. Bloomsbury Publishing; 2012.
10. Stein HH: Combat veterans: impressions of an analytic observer in a non-analytic setting. *J Am Acad Psychoanal Dyn Psychiatr* 2007; 35(4): 575–89.
11. Ben-Shalom U, Weinstein M, Keren T: Studying the eye of the storm: How to study soldiers' effectiveness in irregular warfare. *Mil Psychol-IDF* 2008; 6: 169–208.
12. Glicksohn J, Bozna M: Developing a personality profile of the bomb-disposal expert: the role of sensation seeking and field dependence–independence. *Pers Individ Differ* 2000; 28(1): 85–92.
13. Glicksohn J, Naor-Ziv R: Personality profiling of pilots: traits and cognitive style. *Int J Pers Psychol* 2016; 2(1): 7–14.
14. Ausburn LJ, Brown D: Learning strategy patterns and instructional preferences of career and technical education students. *J STEM Teach Educ* 2006; 43(4): 4.
15. Hayes J, Allinson CW: Cognitive style and the theory and practice of individual and collective learning in organizations. *Hum Relat* 1998; 51(7): 847–71.
16. Riding R, Cheema I: Cognitive styles—an overview and integration. *Educ Psychol* 1991; 11(3–4): 193–215.
17. Witkin HA, Goodenough DR: Field dependence and interpersonal behavior. *Psychol bulletin* 1977; 84(4): 661.
18. Peters M, Laeng B, Latham K, Jackson M, Zaiyouna R, Richardson C: A redrawn Vandenberg and Kuse mental rotations test—different versions and factors that affect performance. *Brain and cogn* 1995; 28(1): 39–58.
19. Boccia M, Vecchione F, Piccardi L, Guariglia C: Effect of cognitive style on learning and retrieval of navigational environments. *Front Pharmacol* 2017; 8: 496.
20. Carter BJ, Cammermeyer M: Biopsychological responses of medical unit personnel wearing chemical defense ensemble in a simulated chemical warfare environment. *Mil Med* 1985; 150(5): 239–49.
21. Moes GS, Lall R, Johnson WB: Personality characteristics of successful navy submarine personnel. *Mil Med* 1996; 161(4): 239–42.
22. Lacoursiere R: Military trauma's unusual appearance. *J Am Acad Psychiatry Law* 2017; 45(2): 221–2.
23. Charmandari E, Tsigos C, Chrousos G: Endocrinology of the stress response. *Annu Rev Physiol* 2005; 67: 259–84.
24. Shamburger T: *Stress and Job Performance of Aeromedical Crewmembers: The Roles of Salivary Cortisol and Social Support*. University of Alabama at Birmingham, Graduate School; 2012.
25. Kirschbaum C, Hellhammer DH: Salivary cortisol in psychobiological research: an overview. *Neuropsychobiology* 1989; 22(3): 150–69.
26. Redondo-Flórez L, Tornero-Aguilera JF, Clemente-Suárez VJ: Could academic experience modulate psychophysiological stress response of biomedical sciences students in laboratory? *Physiol Behav* 2020; 223: 113017.

27. Reini SA: Hypercortisolism as a potential concern for submariners. *Aviat Space Environ Med* 2010; 81(12): 1114–22.
28. Huang G-B, Ma H, Gan X, et al: Circadian misalignment leads to changes in cortisol rhythms, blood biochemical variables and serum miRNA profiles. *Biochem Biophys Res Commun* 2021; 567: 9–16.
29. Martin K, Périard J, Rattray B, Pyne DB: Physiological factors which influence cognitive performance in military personnel. *Hum Factors* 2020; 62(1): 93–123.
30. Ben-Shalom U, Klar Y, Benbenisty Y: 17 Characteristics of Sense-Making in Combat. In: Lawrence J, Matthews MD, (eds). *The Oxford Handbook of Military Psychology*. Oxford University Press 2012: 218.
31. First MB, Spitzer RL, Gibbon M, Williams JB: The structured clinical interview for DSM-III-R personality disorders (SCID-II). Part I: Description. *J Pers disorders* 1995; 9(2): 83–91.
32. Skolnick RB: *Effects of mindfulness-based body scan instruction on anxiety and fear tolerance during in vivo exposure procedure for claustrophobia*. Hofstra University 2014.
33. Van Wijk CH: Personality profiles of divers: integrating results across studies. *Int marit health* 2018; 69(4): 297–303.
34. Balters S, Geeseman JW, Tveten A-K, Hildre HP, Ju W, Steinert M: Mayday, Mayday, Mayday: using salivary cortisol to detect distress (and eustress!) in critical incident training. *Int J Ind Ergon* 2020; 78: 102975.